III B.TECH – I SEMESTER

Electrical Safety Practices And Standards

Open Elective-I | JNTUA R23 · B.Tech CSE

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ELECTRICAL SAFETY PRACTICES AND STANDARDS

(Open Elective-I)

Course Outcomes:

CO1: Understanding the Fundamentals of Electrical Safety -L2

CO2: Identifying and Applying Safety Components -L3

CO3: Analyzing Grounding Practices and Electrical Bonding

CO4: Applying Safety Practices in Electrical Installations and Environments- L4

CO5: Evaluating Electrical Safety Standards and Regulatory Compliance -L5

UNIT I

Introduction To Electrical Safety:

Fundamentals of Electrical safety-Electric Shock- physiological effects of electric current - Safety requirements –Hazards of electricity- Arc - Blast- Causes for electrical failure.

UNIT II

Safety Components:

Introduction to conductors and insulators- voltage classification -safety against over voltages- safety against static electricity-Electrical safety equipment's - Fire extinguishers for electrical safety.

UNIT III

Grounding:

General requirements for grounding and bonding- Definitions- System grounding-Equipment grounding - The Earth - Earthing practices- Determining safe approach distance-Determining arc hazard category.

UNIT IV

Safety Practices:

General first aid- Safety in handling hand held electrical appliances tools- Electrical safety in train stations-swimming pools, external lighting installations, medical locations-Case studies.

UNIT V

Standards For Electrical Safety:

Electricity Acts- Rules & regulations- Electrical standards-NFPA 70 E-OSHA standards-IEEE standards-National Electrical Code 2005 – National Electric Safety code NESC-Statutory requirements from electrical inspectorate

TEXT BOOKS:

- Massimo A.G.Mitolo, "Electrical Safety of Low-Voltage Systems", McGraw Hill, USA, 2009.
- Mohamed El-Sharkawi, "Electric Safety - Practice and Standards", CRC Press, USA, 2014

REFERENCES:

- Kenneth G.Mastrullo, Ray A. Jones, "The Electrical Safety Program Book", Jones and Bartlett Publishers, London, 2nd Edition, 2011.
- Palmer Hickman, "Electrical Safety-Related Work Practices", Jones & Bartlett Publishers, London, 2009.
- Fordham Cooper, W., "Electrical Safety Engineering", Butterworth and Company, London, 1986.
- John Cadick, Mary Capelli-Schellpfeffer, Dennis K. Neitzel, "Electrical Safety Hand book, McGraw-Hill, New York, USA, 4th edition, 2012.